

Practical-01

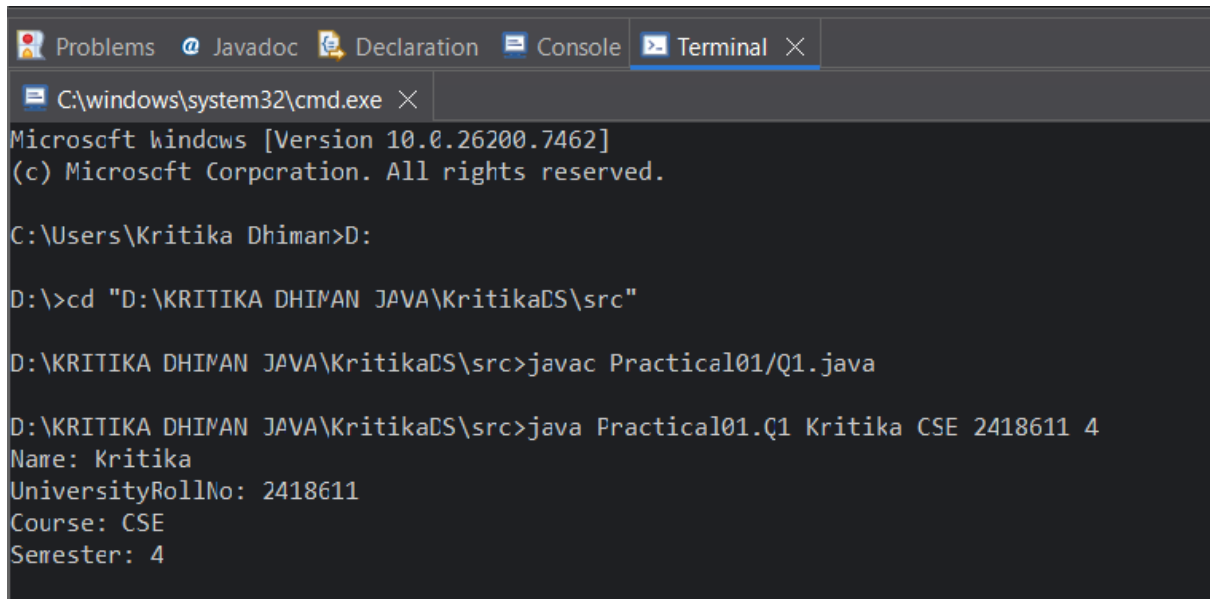
```
package Practical01;
class Q1 {
    public static void main(String[] args) {

        if (args.length < 4) {
            System.out.println("Run like this:");
            System.out.println("java Q1 Name Course RollNo Semester");
            return;
        }

        String name = args[0];
        String course = args[1];
        int rollNo = Integer.parseInt(args[2]);
        int semester = Integer.parseInt(args[3]);

        System.out.println("Name: " + name);
        System.out.println("UniversityRollNo: " + rollNo);
        System.out.println("Course: " + course);
        System.out.println("Semester: " + semester);
    }
}
```

Output



The image shows a screenshot of an IDE's terminal window. The terminal has tabs for 'Problems', 'Javadoc', 'Declaration', 'Console', and 'Terminal'. The 'Terminal' tab is active, showing a Windows command prompt session. The user navigates to the directory 'D:\KRITIKA DHIMAN JAVA\KritikaDS\src' and runs 'javac Practical01/Q1.java'. Then, they run 'java Practical01.Q1 Kritika CSE 2418611 4', which produces the output: 'Name: Kritika', 'UniversityRollNo: 2418611', 'Course: CSE', and 'Semester: 4'.

```
Microsoft Windows [Version 10.0.26200.7462]
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C:\Users\Kritika Dhiman>D:

D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q1.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q1 Kritika CSE 2418611 4
Name: Kritika
UniversityRollNo: 2418611
Course: CSE
Semester: 4
```

Practical-02

```
package Practical01;
import java.util.Scanner;
public class Q2{
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);

        //Displaying Menu
        System.out.println("Select Deposit Type:");
        System.out.println("i.Term Deposit:");
        System.out.println("ii.Recurring Deposit :");
        System.out.println("Enter choice (1 or 2):");
        int choice = sc.nextInt();
        double p,r,a;
        int n;
        switch(choice) {
            case 1:
                //term deposit
                System.out.print("enter principle value:");
                p=sc.nextDouble();
                System.out.print("enter the rate (r):");
                r = sc.nextDouble();
                System.out.print("enter the time in years (n):");
                n = sc.nextInt();

                //formula :  $a=p*[1+r/100]^n$ 
                a=p*Math.pow((1+r/100), n);
                System.out.println("Maturity amount (a) = "+a);
                break;
            case 2:
                //Accept details for recurring deposit
                System.out.print("enter monthyl installment (p): ");
                p=sc.nextDouble();
                System.out.print("ente rate of interest (r): ");
                r=sc.nextDouble();
                System.out.print("enter time period (n): ");
                n=sc.nextInt();

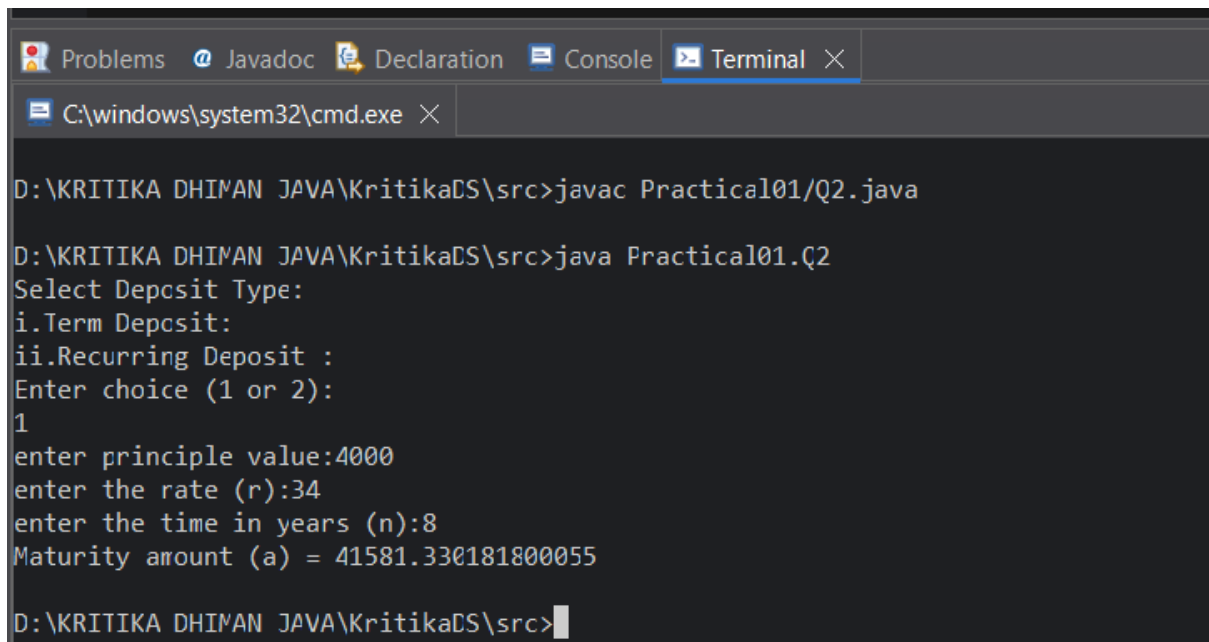
                //formula :  $a=p*n+p*n*(n+1)/2*r/100*1/12$ 

                a=p*n+p*n*(n+1)/2.0*r/100.0*1.0/12.0;
                System.out.println("maturity amount (a)= "+a);
        }
    }
}
```

```
        break;
    default:
        //error message for incorrect choice

        System.out.println("error: invalid option selected ");
    }
    sc.close();
}
```

Output



The screenshot shows an IDE window with a terminal tab active. The terminal title bar indicates the path is C:\windows\system32\cmd.exe. The terminal content shows the following sequence of commands and output:

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q2.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q2
Select Depcsit Type:
i.Term Depcsit:
ii.Recurring Deposit :
Enter choice (1 or 2):
1
enter principle value:4000
enter the rate (r):34
enter the time in years (n):8
Maturity arount (a) = 41581.330181800055

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-03

```
package Practical01;

import java.util.Scanner;

public class Q3{

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input the numbers
        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();
        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();

        // Initialize sum of divisors
        int sum1 = 0, sum2 = 0;

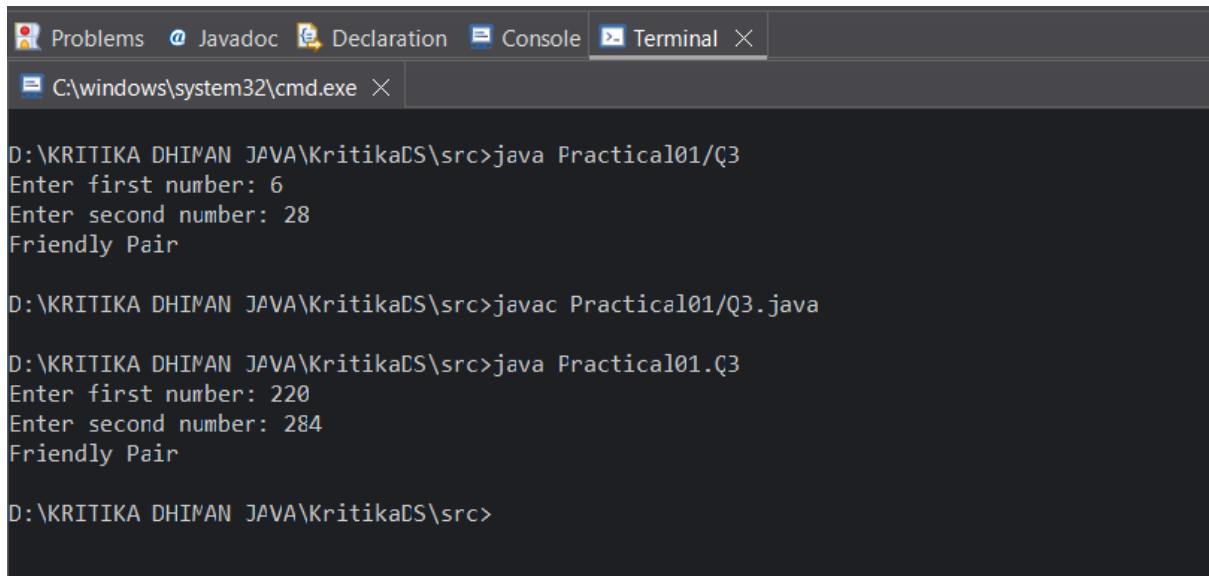
        // Calculate sum of divisors of num1
        for (int i = 1; i < num1/2; i++) {
            if (num1 % i == 0) {
                sum1 += i;
            }
        }

        // Calculate sum of divisors of num2
        for (int i = 1; i < num2/2; i++) {
            if (num2 % i == 0) {
                sum2 += i;
            }
        }

        // Check if they are friendly pair
        if ((sum1 / num1) == (sum2 / num2)) {
            System.out.println("Friendly Pair");
        } else {
            System.out.println("Not Friendly Pair");
        }

        sc.close();
    }
}
```

Output



The screenshot shows an IDE window with a terminal tab active. The terminal displays the execution of a Java program. The first run uses inputs 6 and 28, resulting in 'Friendly Pair'. The second run uses inputs 220 and 284, also resulting in 'Friendly Pair'.

```
Problems Javadoc Declaration Console Terminal X
C:\windows\system32\cmd.exe X

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01/Q3
Enter first number: 6
Enter second number: 28
Friendly Pair

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q3.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q3
Enter first number: 220
Enter second number: 284
Friendly Pair

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-04

```
package Practical01;

import java.util.Scanner;

public class Q4{

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

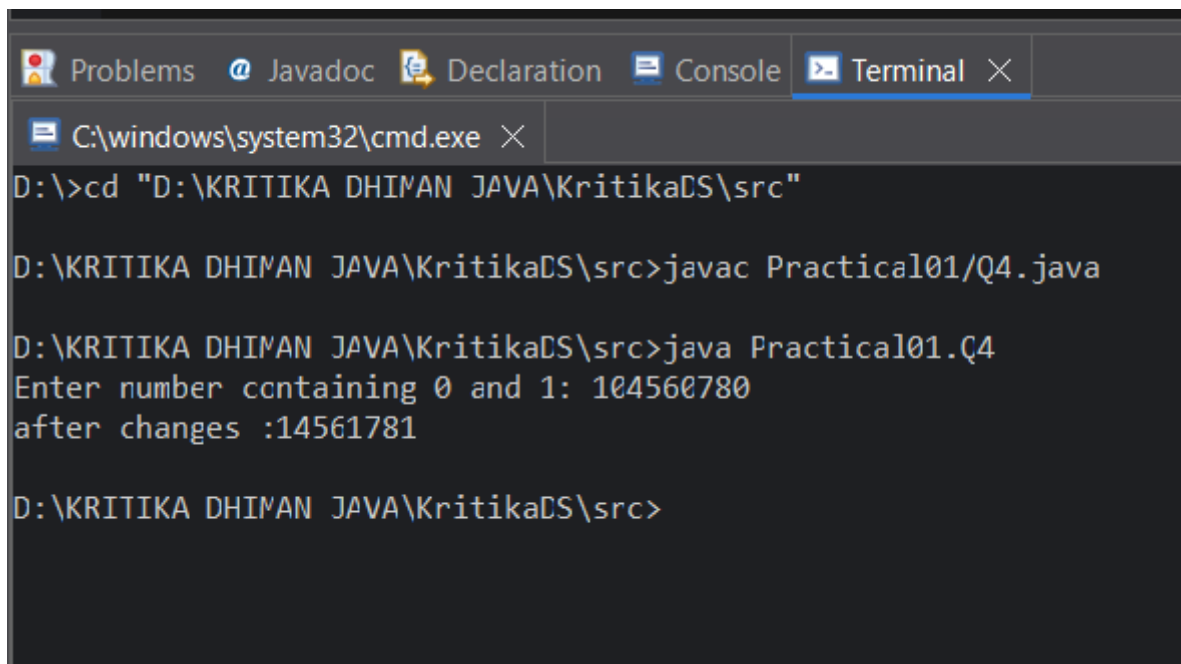
        int num,digit;
        int result = 0;
        int place = 1;

        System.out.print("Enter number containing 0 and 1: ");
        num = sc.nextInt();

        while(num>0)
        {
            digit=num%10;
            if(digit==0) {
                digit=1;
            }else if(digit==1){
                digit=0;
            }

            result=result + digit *place;
            place=place*10;
            num=num/10;
        }
        System.out.println("after changes :"+result);
        sc.close();
    }
}
```


Output



The screenshot shows an IDE interface with a terminal window. The terminal title bar indicates it is running 'C:\windows\system32\cmd.exe'. The terminal content shows the following sequence of commands and output:

```
D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q4.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q4
Enter number containing 0 and 1: 104560780
after changes :14561781

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-05

```
package Practical01;

import java.util.Scanner;

public class Q5 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        boolean less = true;

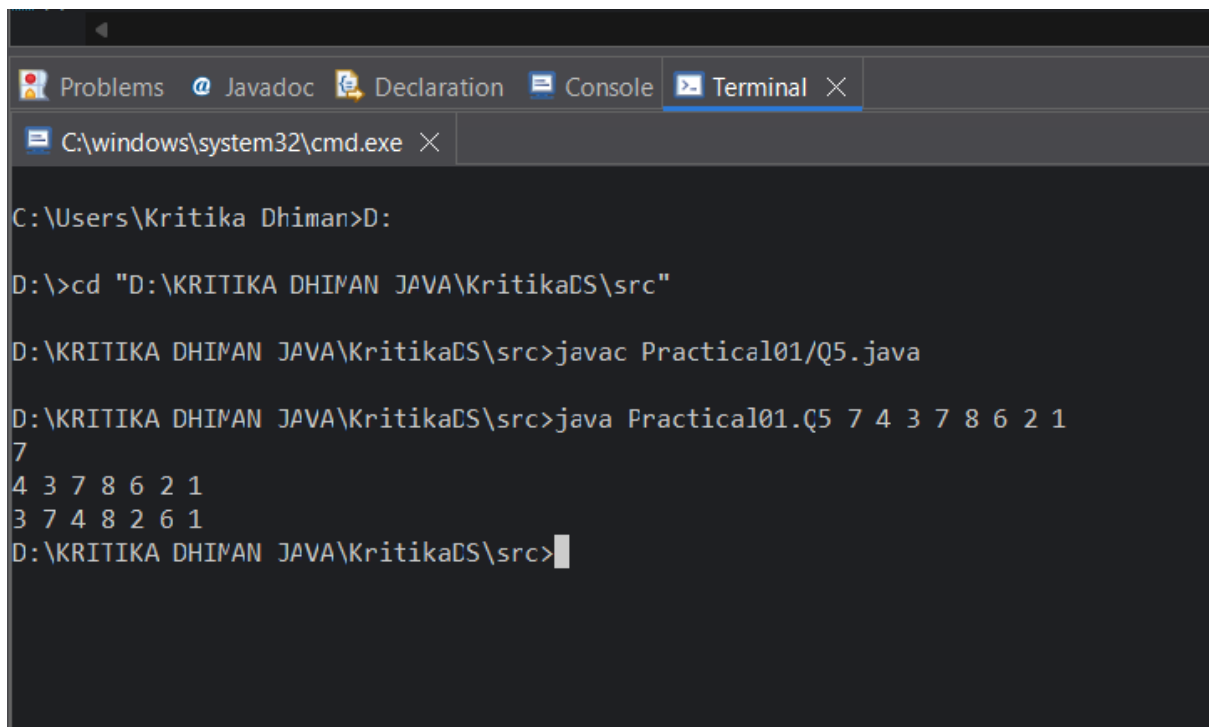
        for (int i = 0; i < n - 1; i++) {
            if (less) {
                if (arr[i] > arr[i + 1]) {
                    int temp = arr[i];
                    arr[i] = arr[i + 1];
                    arr[i + 1] = temp;
                } else {
                    if (arr[i] < arr[i + 1]) {
                        int temp = arr[i];
                        arr[i] = arr[i + 1];
                        arr[i + 1] = temp;
                    }
                }
            }

            less = !less;
        }

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        sc.close();
    }
}
```

Output



The screenshot shows an IDE terminal window with the following content:

```
C:\Users\Kritika Dhiman>D:
D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q5.java
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q5 7 4 3 7 8 6 2 1
7
4 3 7 8 6 2 1
3 7 4 8 2 6 1
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

The terminal window has tabs for Problems, Javadoc, Declaration, Console, and Terminal. The current tab is Terminal, and the command prompt is C:\windows\system32\cmd.exe.

Practical-06

```
package Practical01;

import java.util.Scanner;

public class Q6 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

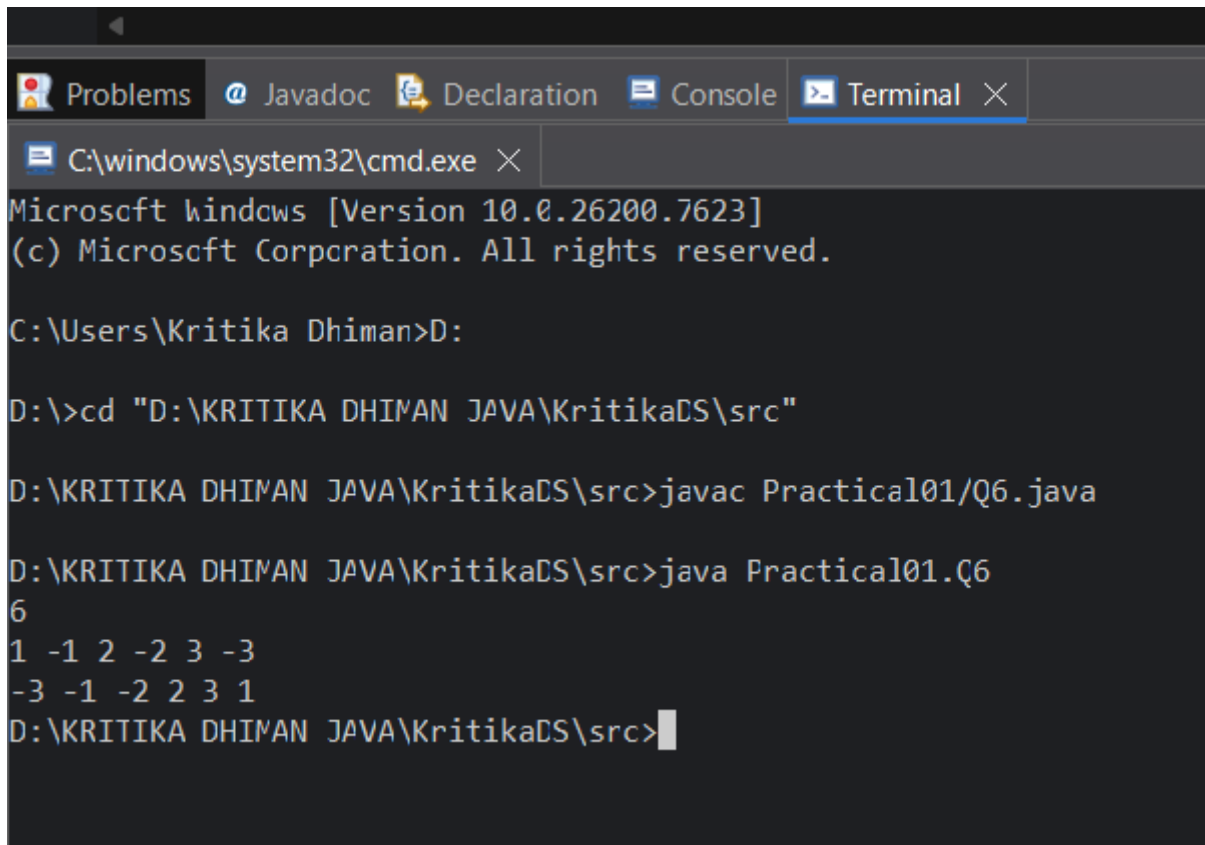
        int left = 0;
        int right = n - 1;

        while (left < right) {

            if (arr[left] < 0) {
                left++;
            }
            else if (arr[right] >= 0) {
                right--;
            }
            else {
                int temp = arr[left];
                arr[left] = arr[right];
                arr[right] = temp;
                left++;
                right--;
            }
        }

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }sc.close();
    }
}
```

Output



The screenshot shows an IDE interface with a terminal window open. The terminal title bar indicates it is running 'C:\windows\system32\cmd.exe'. The terminal output shows the user navigating to the directory 'D:\KRITIKA DHIMAN JAVA\KritikaDS\src' and compiling a file named 'Practical01/Q6.java' using the 'javac' command. After running the 'java' command, the program outputs the number '6' followed by two lines of space-separated integers: '1 -1 2 -2 3 -3' and '-3 -1 -2 2 3 1'.

```
Microsoft Windows [Version 10.0.26200.7623]
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C:\Users\Kritika Dhiman>D:

D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q6.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q6
6
1 -1 2 -2 3 -3
-3 -1 -2 2 3 1
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-06

```
package Practical01;
import java.util.Scanner;

public class Q7 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int row, col;
        System.out.print("Enter number of rows: ");
        row = sc.nextInt();
        System.out.print("Enter number of columns: ");
        col = sc.nextInt();

        int[][] mat = new int[row][col];

        System.out.println("Enter matrix elements:");
        for (int i = 0; i < row; i++) {
            for (int j = 0; j < col; j++) {
                mat[i][j] = sc.nextInt();
            }
        }

        boolean found = false;

        for (int i = 0; i < row; i++) {

            int min = mat[i][0];
            int colIndex = 0;

            for (int j = 1; j < col; j++) {
                if (mat[i][j] < min) {
                    min = mat[i][j];
                    colIndex = j;
                }
            }

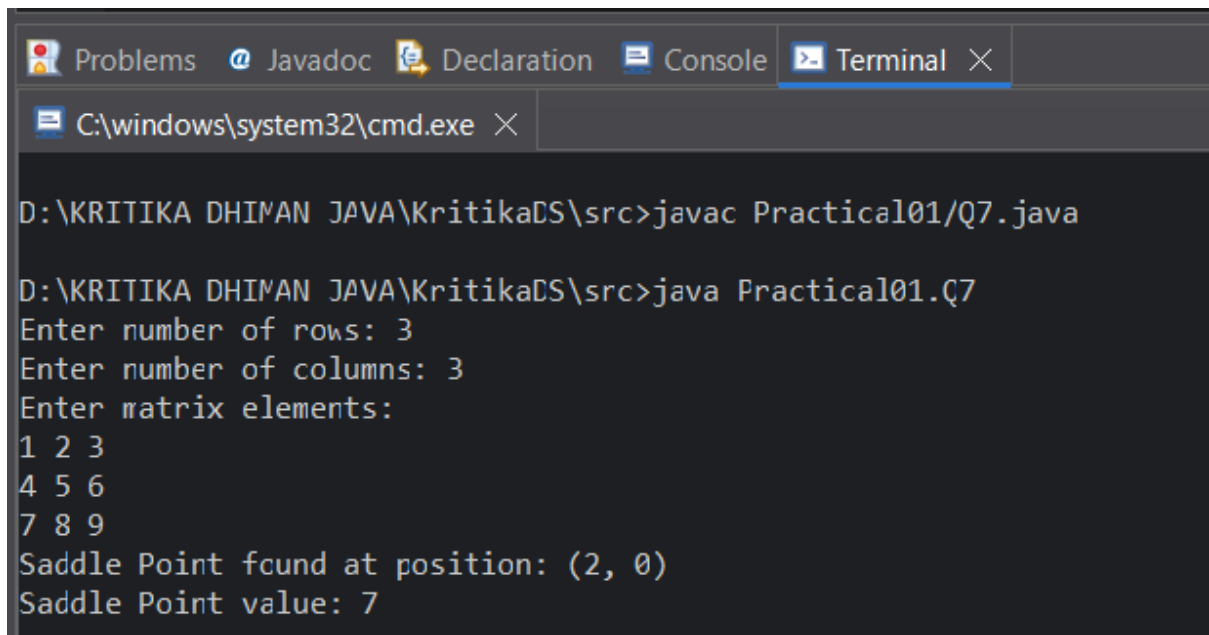
            int k;
            for (k = 0; k < row; k++) {
                if (mat[k][colIndex] > min)
```

```
        break;
    }

    if (k == row) {
        System.out.println("Saddle Point found at position: (" + i + ", " + colIndex + ")");
        System.out.println("Saddle Point value: " + min);
        found = true;
    }
}

if (!found) {
    System.out.println("No Saddle Point found");
}sc.close();
}
```

Output



The screenshot shows an IDE window with a 'Terminal' tab active. The terminal title bar indicates the command prompt is running at 'C:\windows\system32\cmd.exe'. The command prompt shows the following sequence of commands and output:

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q7.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q7
Enter number of rows: 3
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9
Saddle Point found at position: (2, 0)
Saddle Point value: 7
```


Practical-07

```
package Practical01;
import java.util.Scanner;

public class Q7 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int row, col;
        System.out.print("Enter number of rows: ");
        row = sc.nextInt();
        System.out.print("Enter number of columns: ");
        col = sc.nextInt();

        int[][] mat = new int[row][col];

        System.out.println("Enter matrix elements:");
        for (int i = 0; i < row; i++) {
            for (int j = 0; j < col; j++) {
                mat[i][j] = sc.nextInt();
            }
        }

        boolean found = false;

        for (int i = 0; i < row; i++) {

            int min = mat[i][0];
            int colIndex = 0;

            for (int j = 1; j < col; j++) {
                if (mat[i][j] < min) {
                    min = mat[i][j];
                    colIndex = j;
                }
            }

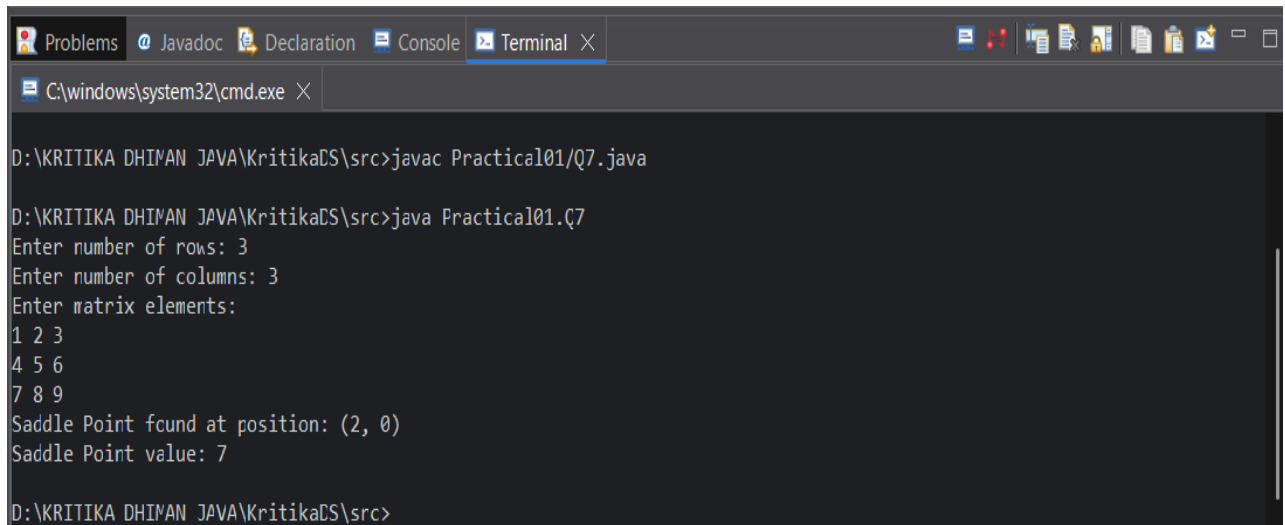
            int k;
            for (k = 0; k < row; k++) {
                if (mat[k][colIndex] > min)
```

```
        break;
    }

    if (k == row) {
        System.out.println("Saddle Point found at position: (" + i + ", " + colIndex + ")");
        System.out.println("Saddle Point value: " + min);
        found = true;
    }
}

if (!found) {
    System.out.println("No Saddle Point found");
}sc.close();
}
```

Output



The screenshot shows an IDE terminal window with the following content:

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q7.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q7
Enter number of rows: 3
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9
Saddle Point found at position: (2, 0)
Saddle Point value: 7

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

The terminal window has a title bar with tabs for 'Problems', 'Javadoc', 'Declaration', 'Console', and 'Terminal'. The 'Terminal' tab is active. The command prompt shows the execution of a Java program that finds a saddle point in a 3x3 matrix.

Practical-08

```
package Practical01;

import java.util.Scanner;

class Bank {

    private String name;
    private String address;
    private int accountNumber;
    private double balance;

    private static int accountCounter = 1000; // starts from 1000

    // Constructor
    public Bank(String name, String address, double balance) {
        this.name = name;
        this.address = address;
        this.balance = balance;
        accountCounter++;
        this.accountNumber = accountCounter; // 1001, 1002, 1003...
    }

    // Display information
    public void display() {
        System.out.println("\n--- Account Details ---");
        System.out.println("Name: " + name);
        System.out.println("Address: " + address);
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    // Deposit
    public void deposit(double amount) {
        balance += amount;
        System.out.println("Amount Deposited Successfully.");
    }

    // Withdraw
    public void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Amount Withdrawn Successfully.");
        }
    }
}
```

```

        } else {
            System.out.println("Insufficient Balance.");
        }
    }

    // Change address
    public void changeAddress(String newAddress) {
        address = newAddress;
        System.out.println("Address Updated Successfully.");
    }

    public int getAccountNumber() {
        return accountNumber;
    }
}

public class Q8 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of depositors: ");
        int n = sc.nextInt();
        sc.nextLine();

        Bank[] accounts = new Bank[n];

        // Creating accounts
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter details for depositor " + (i + 1));

            System.out.print("Name: ");
            String name = sc.nextLine();

            System.out.print("Address: ");
            String address = sc.nextLine();

            System.out.print("Initial Balance: ");
            double balance = sc.nextDouble();
            sc.nextLine();

            accounts[i] = new Bank(name, address, balance);
        }
    }
}

```

```

int choice;

do {
    System.out.println("\n1. Display Account");
    System.out.println("2. Deposit");
    System.out.println("3. Withdraw");
    System.out.println("4. Change Address");
    System.out.println("5. Exit");

    System.out.print("Enter choice: ");
    choice = sc.nextInt();

    if (choice == 5) break;

    System.out.print("Enter Account Number: ");
    int accNo = sc.nextInt();
    sc.nextLine();

    Bank selected = null;

    for (Bank b : accounts) {
        if (b.getAccountNumber() == accNo) {
            selected = b;
            break;
        }
    }

    if (selected == null) {
        System.out.println("Account not found.");
        continue;
    }

    switch (choice) {

        case 1:
            selected.display();
            break;

        case 2:
            System.out.print("Enter amount to deposit: ");
            double dep = sc.nextDouble();
            selected.deposit(dep);
            selected.display();

```

```
        break;

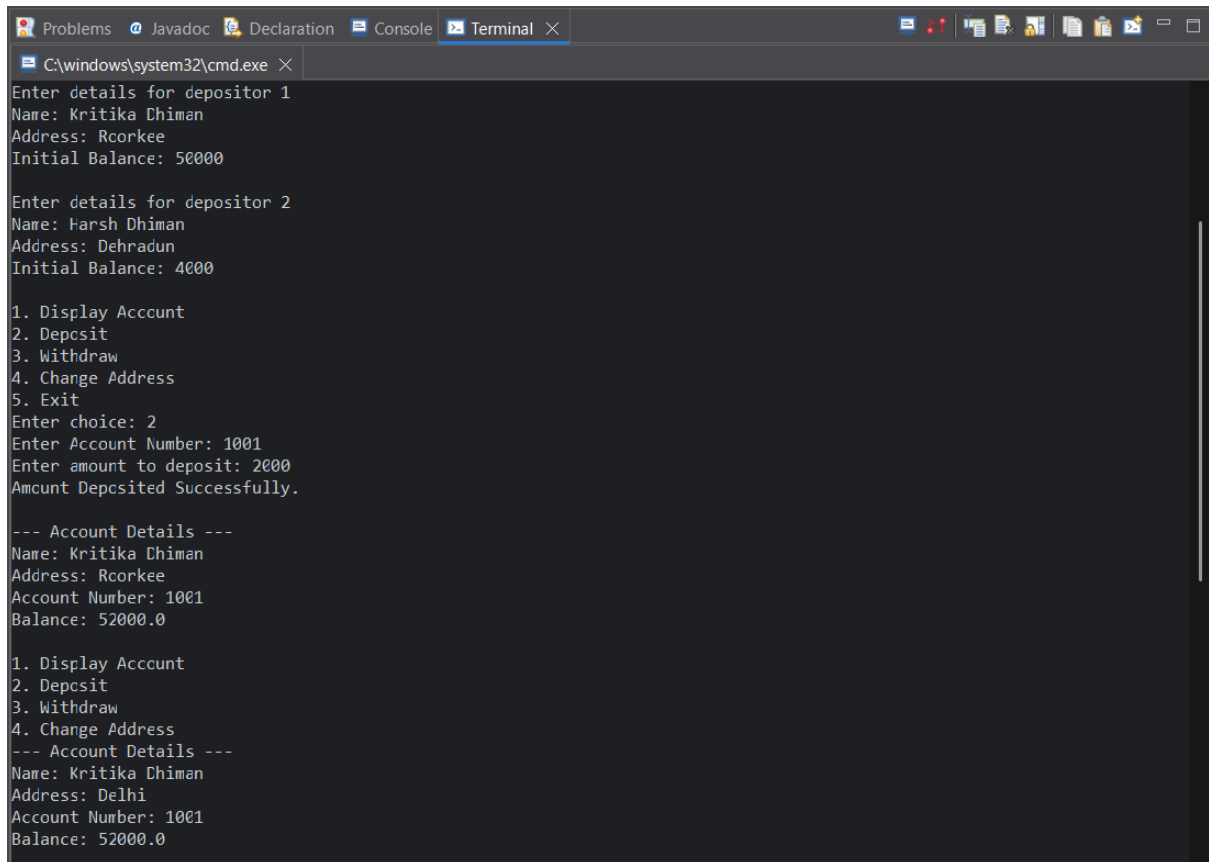
    case 3:
        System.out.print("Enter amount to withdraw: ");
        double wd = sc.nextDouble();
        selected.withdraw(wd);
        selected.display();
        break;

    case 4:
        System.out.print("Enter new address: ");
        String newAddress = sc.nextLine();
        selected.changeAddress(newAddress);
        selected.display();
        break;
    }

    } while (true);

    sc.close();
}
}
```

Output



```
C:\windows\system32\cmd.exe X
Enter details for depositor 1
Name: Kritika Chiman
Address: Roorkee
Initial Balance: 50000

Enter details for depositor 2
Name: Harsh Dhiman
Address: Dehradun
Initial Balance: 4000

1. Display Account
2. Deposit
3. Withdraw
4. Change Address
5. Exit
Enter choice: 2
Enter Account Number: 1001
Enter amount to deposit: 2000
Amount Deposited Successfully.

--- Account Details ---
Name: Kritika Chiman
Address: Roorkee
Account Number: 1001
Balance: 52000.0

1. Display Account
2. Deposit
3. Withdraw
4. Change Address
--- Account Details ---
Name: Kritika Chiman
Address: Delhi
Account Number: 1001
Balance: 52000.0
```


Practical-09

```
package Practical01;

import java.util.Scanner;

class WordExample {

    private String strdata;

    // Parameterized Constructor
    WordExample(String str) {
        strdata = str.trim();
    }

    // Method to check vowel
    private boolean isVowel(char ch) {
        return (ch == 'A' || ch == 'E' || ch == 'I' ||
            ch == 'O' || ch == 'U');
    }

    // Count words beginning and ending with vowel
    void countWord() {

        String sentence = strdata.substring(0, strdata.length() - 1);
        String[] words = sentence.split("\\s+");

        int count = 0;

        for (String word : words) {
            if (word.length() > 0 &&
                isVowel(word.charAt(0)) &&
                isVowel(word.charAt(word.length() - 1))) {
                count++;
            }
        }

        System.out.println("Number of words beginning and ending with vowel: " + count);
    }

    // Rearrange words
    void placeWord() {
```

```

String sentence = strdata.substring(0, strdata.length() - 1);
String[] words = sentence.split("\\s+");

String vowelWords = "";
String otherWords = "";

for (String word : words) {
    if (word.length() > 0 &&
        isVowel(word.charAt(0)) &&
        isVowel(word.charAt(word.length() - 1))) {

        vowelWords += word + " ";
    } else {
        otherWords += word + " ";
    }
}

String result = vowelWords + otherWords;

System.out.println("Rearranged Sentence:");
System.out.println(result.trim());
}
}

public class Q9 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a sentence in UPPER CASE ending with ., ? or !");
        String input = sc.nextLine();

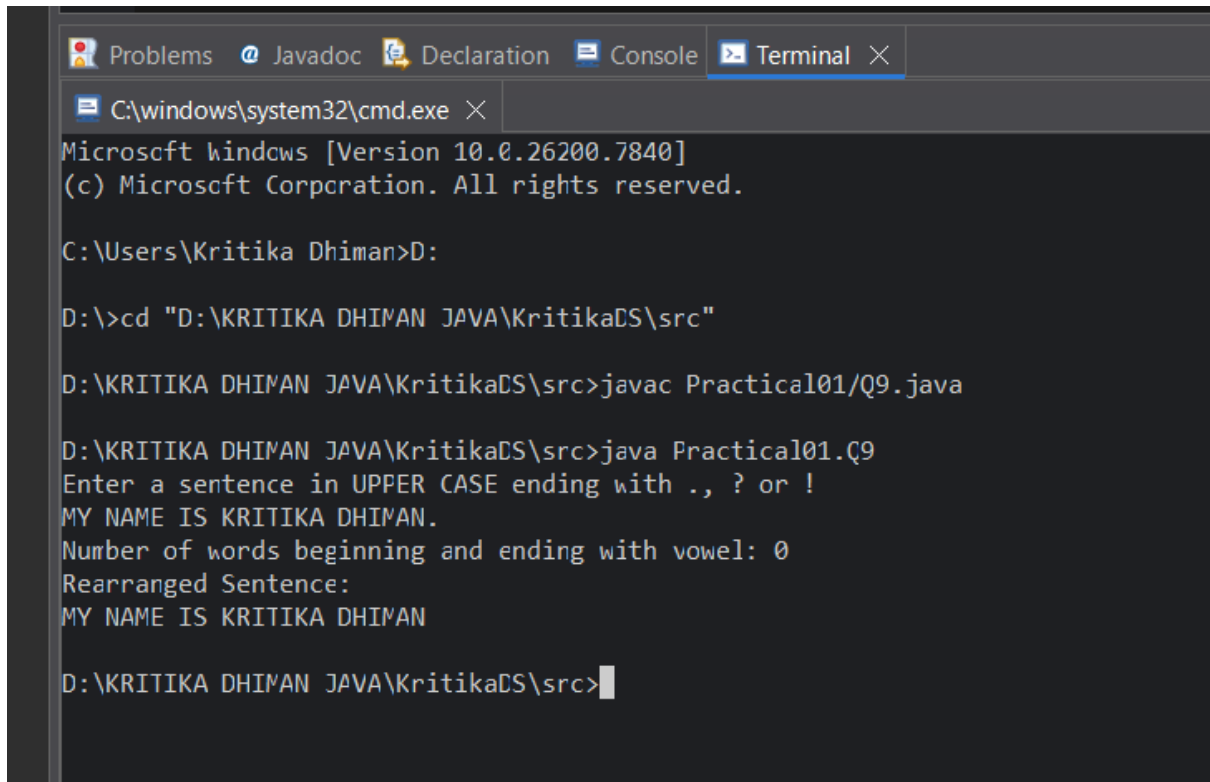
        WordExample obj = new WordExample(input);

        obj.countWord();
        obj.placeWord();

        sc.close();
    }
}

```

Output



The screenshot shows a Java IDE with a terminal window open. The terminal title bar indicates the path `C:\windows\system32\cmd.exe`. The terminal output shows the following sequence of commands and results:

```
Microsoft Windows [Version 10.0.26200.7840]
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C:\Users\Kritika Dhiman>D:

D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q9.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q9
Enter a sentence in UPPER CASE ending with ., ? or !
MY NAME IS KRITIKA DHIMAN.
Number of words beginning and ending with vowel: 0
Rearranged Sentence:
MY NAME IS KRITIKA DHIMAN

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-10

```
package Practical01;
```

```
class ArrayDemo {
```

```
    void arrayFunc(int arr[], int target) {  
        System.out.println("Pairs of elements whose sum is " + target + " are :");  
        for(int i = 0; i < arr.length; i++) {  
            for(int j = i + 1; j < arr.length; j++) {  
                if(arr[i] + arr[j] == target) {  
                    System.out.println(arr[i] + " + " + arr[j] + " = " + target);  
                }  
            }  
        }  
    }  
}
```

```
void arrayFunc(int A[], int B[], int p, int q) {  
    int temp[] = new int[p + q];  
    int i = 0, j = 0, k = 0;
```

```
    while(i < p && j < q) {  
        if(A[i] < B[j]) {  
            temp[k] = A[i];  
            i++;  
        } else {  
            temp[k] = B[j];  
            j++;  
        }  
        k++;  
    }
```

```
    while(i < p) {  
        temp[k] = A[i];  
        i++;  
        k++;  
    }
```

```
    while(j < q) {  
        temp[k] = B[j];  
        j++;  
        k++;  
    }
```

```

        for(i = 0; i < p; i++) {
            A[i] = temp[i];
        }

        for(j = 0; j < q; j++) {
            B[j] = temp[p + j];
        }

        System.out.println("Sorted Arrays:");
        System.out.print("A: ");
        for(i = 0; i < p; i++) {
            System.out.print(A[i] + " ");
        }

        System.out.println();
        System.out.print("B: ");
        for(j = 0; j < q; j++) {
            System.out.print(B[j] + " ");
        }
    }
}

public class Q10 {
    public static void main(String args[]) {

        ArrayDemo obj = new ArrayDemo();

        int numbers[] = {4, 6, 5, -10, 8, 5, 20};
        obj.arrayFunc(numbers, 10);

        System.out.println();

        int A[] = {1, 5, 6, 7, 8, 10};
        int B[] = {2, 4, 9};

        obj.arrayFunc(A, B, A.length, B.length);
    }
}

```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q10.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q10
Pairs of elements whose sum is 10 are :
4 + 6 = 10
5 + 5 = 10
-10 + 20 = 10

Sorted Arrays:
A: 1 2 4 5 6 7
B: 8 9 10
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-11

```
package Practical01;
```

```
abstract class Shape {
```

```
    abstract void rectangleArea(double length, double breadth);
```

```
    abstract void squareArea(double side);
```

```
    abstract void circleArea(double radius);
```

```
}
```

```
class Area extends Shape {
```

```
    void rectangleArea(double length, double breadth) {
```

```
        System.out.println("Area of Rectangle: " + (length * breadth));
```

```
    }
```

```
    void squareArea(double side) {
```

```
        System.out.println("Area of Square: " + (side * side));
```

```
    }
```

```
    void circleArea(double radius) {
```

```
        System.out.println("Area of Circle: " + (Math.PI * radius * radius));
```

```
    }
```

```
}
```

```
// Main class
```

```
public class Q11 {
```

```
    public static void main(String[] args) {
```

```
        Shape obj = new Area(); //  No error now
```

```
        obj.rectangleArea(10, 5);
```

```
        obj.squareArea(4);
```

```
        obj.circleArea(3);
```

```
    }
```

```
}
```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q11.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q11
Area of Rectangle: 50.0
Area of Square: 16.0
Area of Circle: 28.274333882308138

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```


Practical-12

```
package Practical01;

import java.util.Scanner;

abstract class Temperature {

    double temp;

    void setTempData(double t) {
        temp = t;
    }

    abstract void changeTemp();
}

class Fahrenheit extends Temperature {

    double ctemp;

    void changeTemp() {
        ctemp = (5.0 / 9.0) * (temp - 32);
        System.out.println("Temperature in Celsius: " + ctemp);
    }
}

class Celsius extends Temperature {

    double ftemp;

    void changeTemp() {
        ftemp = (9.0 / 5.0) * temp + 32;
        System.out.println("Temperature in Fahrenheit: " + ftemp);
    }
}

public class Q12{

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.println("Enter temperature in Fahrenheit:");
double f = sc.nextDouble();

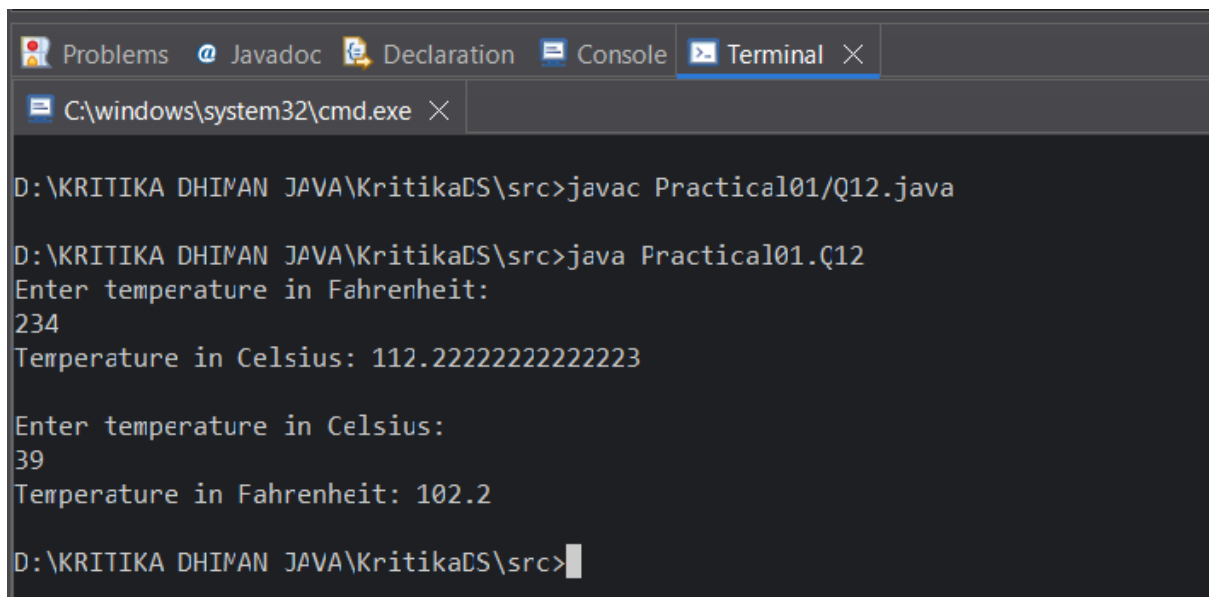
Temperature obj1 = new Fahrenheit();
obj1.setTempData(f);
obj1.changeTemp();

System.out.println("\nEnter temperature in Celsius:");
double c = sc.nextDouble();

Temperature obj2 = new Celsius();
obj2.setTempData(c);
obj2.changeTemp();

sc.close();
}
}
```

Output



The screenshot shows an IDE window with a terminal tab active. The terminal title bar indicates the path C:\windows\system32\cmd.exe. The terminal content shows the execution of a Java program. The user runs 'javac Practical01/Q12.java' and then 'java Practical01.Q12'. The program prompts for temperature in Fahrenheit, where '234' is entered, resulting in 'Temperature in Celsius: 112.22222222222223'. It then prompts for temperature in Celsius, where '39' is entered, resulting in 'Temperature in Fahrenheit: 102.2'. The terminal ends with the prompt 'D:\KRITIKA DHIMAN JAVA\KritikaDS\src>'.

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q12.java
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q12
Enter temperature in Fahrenheit:
234
Temperature in Celsius: 112.22222222222223

Enter temperature in Celsius:
39
Temperature in Fahrenheit: 102.2

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-13

```
package Practical01;

import java.util.Scanner;

interface Volume {
    void volume();
}

class Cone implements Volume {
    double r, h;

    Cone(double r, double h) {
        this.r = r;
        this.h = h;
    }

    public void volume() {
        double v = (1.0/3) * 3.14 * r * r * h;
        System.out.println("Volume of Cone = " + v);
    }
}

class Hemisphere implements Volume {
    double r;

    Hemisphere(double r) {
        this.r = r;
    }

    public void volume() {
        double v = (2.0/3) * 3.14 * r * r * r;
        System.out.println("Volume of Hemisphere = " + v);
    }
}

class Cylinder implements Volume {
    double r, h;

    Cylinder(double r, double h) {
        this.r = r;
        this.h = h;
    }
}
```

```

    public void volume() {
        double v = 3.14 * r * r * h;
        System.out.println("Volume of Cylinder = " + v);
    }
}

public class Q13 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius and height of Cone: ");
        double r1 = sc.nextDouble();
        double h1 = sc.nextDouble();
        Volume v1 = new Cone(r1, h1);
        v1.volume();

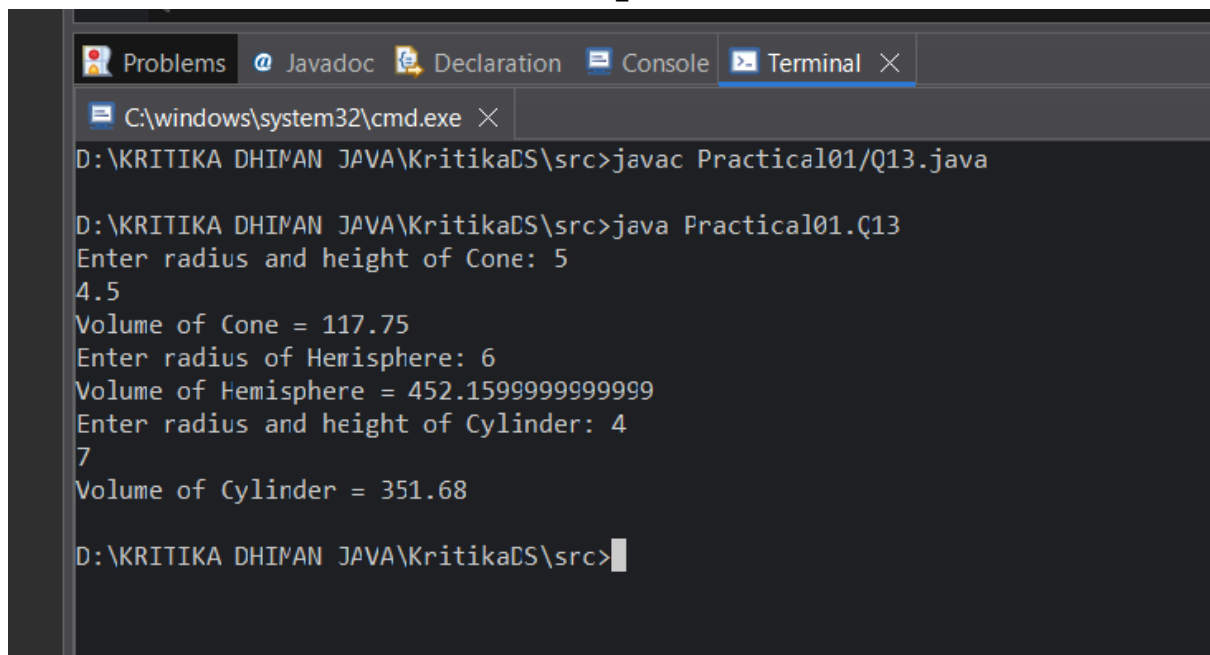
        System.out.print("Enter radius of Hemisphere: ");
        double r2 = sc.nextDouble();
        Volume v2 = new Hemisphere(r2);
        v2.volume();

        System.out.print("Enter radius and height of Cylinder: ");
        double r3 = sc.nextDouble();
        double h3 = sc.nextDouble();
        Volume v3 = new Cylinder(r3, h3);
        v3.volume();

        sc.close();
    }
}

```

Output



```
Problems Javadoc Declaration Console Terminal X
C:\windows\system32\cmd.exe X
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q13.java
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q13
Enter radius and height of Cone: 5
4.5
Volume of Cone = 117.75
Enter radius of Hemisphere: 6
Volume of Hemisphere = 452.1599999999999
Enter radius and height of Cylinder: 4
7
Volume of Cylinder = 351.68
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-14

Student Package :

```
package Student;
public class Student { private int studentId;
    private String name;
    private int age;

    public Student(int studentId, String name, int age) {
        this.studentId=studentId;
        this.name=name;
        this.age=age;
    }public String getstudentinfo() {
        return "StudentID : " + studentId + ", Student Name: " + name + ",
Age: " + age; }
    }
```

Course Package:

```
package Course;
public class Course {private int courseId;
    private String coursename;
    private int credits;
    public Course(int courseId, String coursename, int credits) {
        this.courseId=courseId;
        this.coursename=coursename;
        this.credits=credits;
    }
    public String getcourseinfo() {
        return "CourseID : " + courseId + ", Course Name: " + coursename + ",
Credits: " + credits; }
    }
```

Enrollment Package:

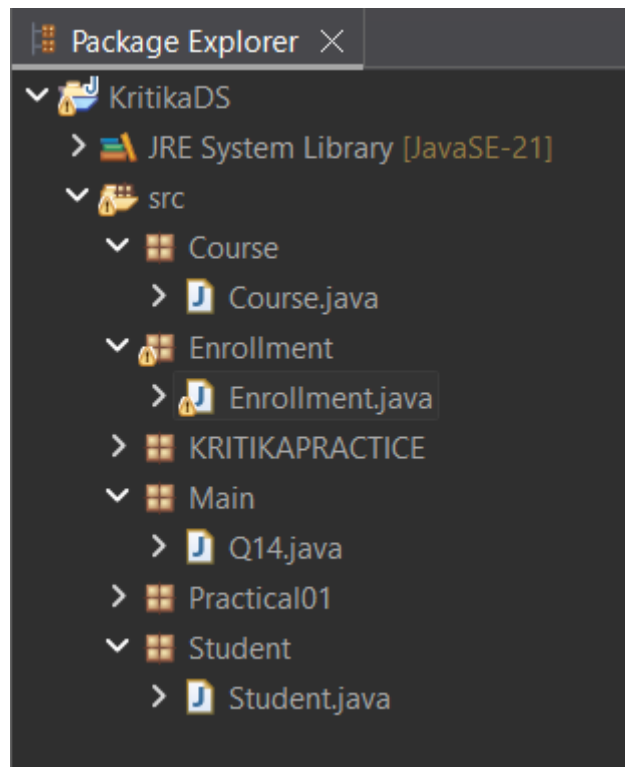
```
package Enrollment;
import Course.Course;
import Student.Student;
public class Enrollment {
    private Student Student;
    private Course Course;
    public Enrollment(Student Student,Course Course) {
        this.Student = Student;
```

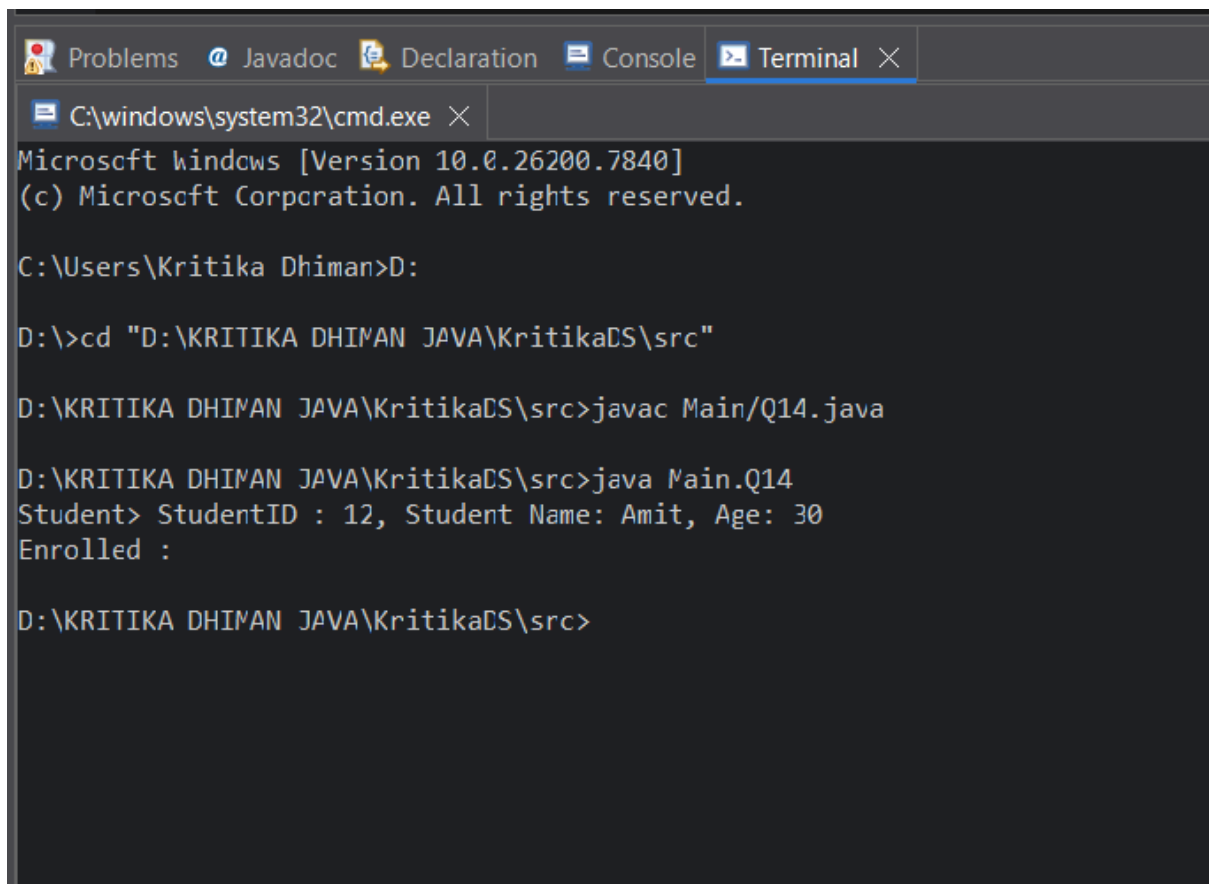
```
        this.Course = Course;}  
    public String getEnrollmentInfo() {return "Student> " + Student.getstudentinfo() +  
"\nEnrolled : ";  
    }}
```

Main:

```
package Main;  
  
import Course.Course;  
import Enrollment.Enrollment;  
import Student.Student;  
  
public class Q14 {  
  
    public static void main(String[] args) {  
        Student objStudent = new Student(12, "Amit", 30);  
        Course objCourse = new Course(1234,"JAVA", 100);  
        Enrollment objEnrollment = new Enrollment(objStudent , objCourse);  
        System.out.println(objEnrollment.getEnrollmentInfo());  
    }  
  
}
```


Output





The image shows a screenshot of an IDE's terminal window. The terminal has tabs for 'Problems', 'Javadoc', 'Declaration', 'Console', and 'Terminal'. The 'Terminal' tab is active, showing a command prompt session. The user has navigated to the directory 'D:\KRITIKA DHIMAN JAVA\KritikaDS\src' and compiled the file 'Main/Q14.java' using 'javac'. Then, they ran the program using 'java Main.Q14', which displayed the output: 'Student> StudentID : 12, Student Name: Amit, Age: 30' followed by 'Enrolled :'. The terminal window title bar indicates it is running 'C:\windows\system32\cmd.exe'.

```
C:\windows\system32\cmd.exe X
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kritika Dhiman>D:

D:\>cd "D:\KRITIKA DHIMAN JAVA\KritikaDS\src"

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Main/Q14.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Main.Q14
Student> StudentID : 12, Student Name: Amit, Age: 30
Enrolled :

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical-15

```
package Practical01;
import java.util.*;

class InvalidDataException extends Exception {
    /**
     *
     */
    private static final long serialVersionUID = 1L;

    public InvalidDataException(String message) {
        super(message);
    }
}

class Employee {
    int empId;
    String name;
    int deptId;

    Employee(int empId, String name, int deptId) throws InvalidDataException {

        // Condition 1: First letter capital
        if (!Character.isUpperCase(name.charAt(0))) {
            throw new InvalidDataException("Name must start with a capital letter");
        }

        // Condition 2: Employee ID range
        if (empId < 2001 || empId > 5001) {
            throw new InvalidDataException("Employee ID must be between 2001 and 5001");
        }

        // Condition 3: Department ID range
        if (deptId < 1 || deptId > 5) {
            throw new InvalidDataException("Department ID must be between 1 and 5");
        }

        this.empId = empId;
        this.name = name;
        this.deptId = deptId;
    }

    void display() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Name: " + name);
        System.out.println("Department ID: " + deptId);
    }
}

public class Q15 {
    public static void main(String[] args) {
        @SuppressWarnings("resource")
```

```
        Scanner sc = new Scanner(System.in);

    try {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();

        System.out.print("Enter Name: ");
        String name = sc.next();

        System.out.print("Enter Department ID: ");
        int dept = sc.nextInt();

        Employee e = new Employee(id, name, dept);
        System.out.println("\nEmployee Details:");
        e.display();

    } catch (InvalidDataException e) {
        System.out.println("Error: " + e.getMessage());
    } catch (Exception e) {
        System.out.println("Error: Invalid input type");
    }
}
```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q15.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q15
Enter Employee ID: 2023
Enter Name: Kritika
Enter Department ID: 3

Employee Details:
Employee ID: 2023
Name: Kritika
Department ID: 3
```

Practical-16

```
package Practical01;

import java.util.Scanner;
class MyCalculator{

    int power(int n,int p) throws Exception{

        if(n==0 & p==0) {
            throw new Exception("n and p should not be zero");
        }

        if(n<0 || p<0) {
            throw new Exception("n and p should non negative ");
        }
        return (int) Math.pow(n,p);
    }
}

public class Q16{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        MyCalculator obj = new MyCalculator();

        while (sc.hasNextInt()) {
            int n = sc.nextInt();
            int p = sc.nextInt();

            try {
                System.out.println(obj.power(n, p));
            } catch (Exception e) {
                System.out.println(e);
            }
        }
        sc.close();
    }
}
```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q16.java

D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q16
3 5
243
2 4
16
0 0
java.lang.Exception: n and p should not be zero
-1 -2
java.lang.Exception: n an p shculd non negative
-1 -3
java.lang.Exception: n an p shculd non negative
```

Practical – 17

```
package Practical01;

import java.io.*;

public class Q17 {

    public static boolean isPalindrome(String word) {
        word = word.toLowerCase();
        int left = 0, right = word.length() - 1;

        while (left < right) {
            if (word.charAt(left) != word.charAt(right)) {
                return false;
            }
            left++;
            right--;
        }
        return true;
    }

    public static void main(String[] args) {
        int count = 0;

        try {
            BufferedReader br = new BufferedReader(new FileReader("C:\\Users\\Kritika
Dhiman\\OneDrive\\Desktop\\myFile.txt"));
            String line;

            while ((line = br.readLine()) != null) {
                String[] words = line.split("\\s+");

                for (String word : words) {

                    word = word.replaceAll("[^a-zA-Z]", "");

                    if (!word.isEmpty() && isPalindrome(word)) {
                        count++;
                    }
                }
            }

            br.close();

            System.out.println("Number of palindromes = " + count);

        } catch (IOException e) {
            System.out.println("Error reading file");
        }
    }
}
```


OUTPUT

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q17.java  
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q17  
Number of palindromes = 4
```

Practical – 18

```
package Practical01;

public class Q18 {
    public static void main(String[] args) {

        CustomThread t1 = new CustomThread("CSTHREAD");
        CustomThread t2 = new CustomThread("ITTHREAD");

        t1.start();
        t2.start();
    }
}

class CustomThread extends Thread {

    public CustomThread(String name) {
        super(name);
    }

    @Override
    public void run() {
        try {
            for (int i = 1; i <= 5; i++) {
                System.out.println(Thread.currentThread().getName());
                Thread.sleep(500);
            }
        } catch (InterruptedException e) {
            System.out.println("Thread interrupted: " + Thread.currentThread().getName());
        }
    }
}
```

OUTPUT

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q18.java
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q18
CSTHREAD
ITTHREAD
ITTHREAD
CSTHREAD
ITTHREAD
CSTHREAD
ITTHREAD
CSTHREAD
ITTHREAD
CSTHREAD
```

Practical – 19

```
package Practical01;
import java.util.LinkedList;
import java.util.Queue;
import java.util.Random;

public class Q19 {

    public static void main(String[] args) {

        Queue<Integer> buffer = new LinkedList<>();
        int maxSize = 10;

        Thread producer = new Producer(buffer, maxSize, "PRODUCER");
        Thread consumer = new Consumer(buffer, maxSize, "CONSUMER");

        producer.start();
        consumer.start();

    }

}

class Producer extends Thread {
    private Queue<Integer> queue;
    private int maxSize;

    public Producer(Queue<Integer> queue, int maxSize, String name) {
        super(name);
        this.queue = queue;
        this.maxSize = maxSize;
    }

    @Override
    public synchronized void run() {
        while (true) {
            synchronized (queue) {
                while (queue.size() == maxSize) {
                    try {
                        System.out.println("Queue is full, " + "Producer
thread waiting for "
```

```

        + "consumer to take something
from queue");

        queue.wait();
    } catch (Exception ex) {
        ex.printStackTrace();
    }
}
Random random = new Random();
int i = random.nextInt();
System.out.println("Producing value : " + i);
queue.add(i);
queue.notify();
    }
}

}

class Consumer extends Thread {
    private Queue<Integer> queue;
    private int maxSize;

    public Consumer(Queue<Integer> queue, int maxSize, String name) {
        super(name);
        this.queue = queue;
        this.setMaxSize(maxSize);
    }

    @Override
    public void run() {
        while (true) {
            synchronized (queue) {
                while (queue.isEmpty()) {
                    System.out.println("Queue is empty," + "Consumer
thread is waiting"
                                + " for producer thread to put something
in queue");
                    try {
                        queue.wait();
                    } catch (Exception ex) {
                        ex.printStackTrace();
                    }
                }
            }
            System.out.println("Consuming value : " + queue.remove());

```

```
        queue.notify();
    }
}

public int getMaxSize() {
    return maxSize;
}

public void setMaxSize(int maxSize) {
    this.maxSize = maxSize;
}
}
```

OUTPUT

```
Consuming value : 204103902
Consuming value : -1216954266
Consuming value : 391954822
Consuming value : 615800132
Consuming value : 1091014457
Consuming value : 596263617
Consuming value : 197272197
Consuming value : 1370388174
Consuming value : 495225032
Consuming value : -1977360142
Queue is empty, Consumer thread is waiting for producer thread to put something in queue
Producing value : 181743758
Producing value : -552217158
Producing value : 117230251
Producing value : -1335805266
Producing value : -696273944
Producing value : 479773102
Producing value : 1392629566
Producing value : 1200883322
Producing value : 1846267518
Producing value : 1020024867
Queue is full, Producer thread waiting for consumer to take something from queue
Consuming value : 181743758
Consuming value : -552217158
Consuming value : 117230251
Consuming value : -1335805266
Consuming value : -696273944
Consuming value : 479773102
Consuming value : 1392629566
Consuming value : 1200883322
Consuming value : 1846267518
```

Practical – 20

```
package Practical01;
```

```
import java.util.ArrayList;
public class Q20 {
    //1.By using loop
    public static void removeEvenLength(ArrayList<String> List) {
        for(int i=0;i<List.size();i++) {
            if(List.get(i).length()%2==0) {
                List.remove(i);
                i--; }
        }
    }

    /*Iterator<String> itr = list.iterator();
    while(itr.hasNext ()) {
        String temp = itr.next();
        if(temp.length() % 2 ==0) {
            itr.remove();
        }
    }*/

    public static void main(String[] args) {
        ArrayList<String> words = new ArrayList<>();
        words.add("apple");
        words.add("banana");
        words.add("pear");
        words.add("kiwi");
        removeEvenLength(words);
        System.out.println(words);
    }
}
```


OUTPUT

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q20.java  
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q20  
[apple]  
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>
```

Practical – 21

```
package Practical01;
import java.util.ArrayList;

public class Q21 {
    public static void swapPairs(ArrayList<String> list) {
        for (int i = 0; i < list.size() - 1; i += 2) {
            String temp = list.get(i);
            list.set(i, list.get(i + 1));
            list.set(i + 1, temp);
        }
    }

    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();

        list.add("four");
        list.add("score");
        list.add("and");
        list.add("seven");
        list.add("years");
        list.add("ago");

        swapPairs(list);

        System.out.println(list);
    }
}
```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q21.java  
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q21  
[score, four, seven, and, ago, years]
```

Practical – 22

```
package Practical01;
import java.util.*;
public class Q22 {

    public static List<Integer> alternate(List<Integer> list1, List<Integer> list2) {
        List<Integer> result = new ArrayList<>();

        int i = 0;

        // Add elements alternately
        while (i < list1.size() && i < list2.size()) {
            result.add(list1.get(i));
            result.add(list2.get(i));
            i++;
        }

        // Add remaining elements from list1
        while (i < list1.size()) {
            result.add(list1.get(i));
            i++;
        }

        // Add remaining elements from list2
        while (i < list2.size()) {
            result.add(list2.get(i));
            i++;
        }

        return result;
    }

    public static void main(String[] args) {
        List<Integer> list1 = Arrays.asList(1, 2, 3, 4, 5);
        List<Integer> list2 = Arrays.asList(6, 7, 8, 9, 10, 11, 12);

        List<Integer> result = alternate(list1, list2);

        System.out.println(result);
    }
}
```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>javac Practical01/Q22.java  
D:\KRITIKA DHIMAN JAVA\KritikaDS\src>java Practical01.Q22  
[1, 6, 2, 7, 3, 8, 4, 9, 5, 10, 11, 12]
```

Practical – 23

```
package Practical01;

import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Q23 extends JFrame implements ActionListener {

    private static final long serialVersionUID = 1L;
    JTextField inputField, resultField;
    JButton countBtn, resetBtn, exitBtn;

    public Q23() {
        setTitle("Vowel Counter");
        setSize(350, 200);
        setLayout(new GridLayout(3, 2, 10, 10));

        JLabel inputLabel = new JLabel("Enter String:");
        JLabel resultLabel = new JLabel("Result:");

        inputField = new JTextField();
        resultField = new JTextField();
        resultField.setEditable(false);

        countBtn = new JButton("Count Vowel");
        resetBtn = new JButton("Reset");
        exitBtn = new JButton("Exit");

        countBtn.addActionListener(this);
        resetBtn.addActionListener(this);
        exitBtn.addActionListener(this);

        add(inputLabel);
        add(inputField);
        add(resultLabel);
        add(resultField);
        add(countBtn);
        add(resetBtn);
        add(exitBtn);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == countBtn) {
            String text = inputField.getText().toLowerCase();
            int count = 0;

            for (int i = 0; i < text.length(); i++) {
                char ch = text.charAt(i);
```

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        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            count++;
        }
    }

    resultField.setText(String.valueOf(count));
}

else if (e.getSource() == resetBtn) {
    inputField.setText("");
    resultField.setText("");
}

else if (e.getSource() == exitBtn) {
    System.exit(0);
}
}

public static void main(String[] args) {
    new Q23();
}
}

```

Output

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src> javac Practical01/Q23.java
```

```
D:\KRITIKA DHIMAN JAVA\KritikaDS\src> java Practical01.Q23
```

Vowel Counter

Enter String: KRITIKA Result:

3

CountVowel Reset

Exit